

ASSIGNMENT 6

QUESTION 1

Create the following data set with 20 different years and perform the following operations using „dplyr“ library.

SOLUTION

```
# Question 1: Create the following data set with 20 different years and perform the
following operations using „dplyr“ library.
df <- data.frame(
  country = rep(c("India", "USA", "China", "Brazil", "UK", "Germany", "France", "Italy",
"Canada", "Australia",
                "Japan", "South Korea", "Mexico", "South Africa", "Russia",
                "Spain", "Netherlands", "Sweden", "Norway", "Denmark", "Finland",
"Poland", "Portugal",
                "Argentina", "Chile", "Colombia", "New Zealand", "Thailand", "Vietnam",
"Philippines"), each = 2),
  continent = rep(c("Asia", "North America", "Asia", "South America", "Europe", "Europe",
"Europe", "Europe", "North America", "Oceania",
                "Asia", "Asia", "North America", "Africa", "Europe",
                "Europe", "Europe", "Europe", "Europe", "Europe", "Europe", "Europe",
"Europe",
                "South America", "South America", "South America", "Oceania", "Asia",
"Asia", "Asia"), each = 2),
  year = rep(2000:2014, times = 2),
  lifeExp = runif(60, 60, 85),
  pop = sample(1e6:1.4e9, 60, replace = TRUE),
  gdpPerc = sample(3000:60000, 60, replace = TRUE)
)

print(summary(df))
```

OUTPUT

country	continent	year	lifeExp
Length:60	Length:60	Min. :2000	Min. :60.26
Class :character	Class :character	1st Qu.:2003	1st Qu.:68.36
Mode :character	Mode :character	Median :2007	Median :72.52
		Mean :2007	Mean :73.13
		3rd Qu.:2011	3rd Qu.:78.68
		Max. :2014	Max. :84.47
pop	gdpPerc		
Min. :8.012e+06	Min. : 4276		
1st Qu.:2.126e+08	1st Qu.:14291		
Median :6.008e+08	Median :26167		
Mean :6.287e+08	Mean :28973		
3rd Qu.:1.007e+09	3rd Qu.:41720		
Max. :1.313e+09	Max. :58945		

QUESTION 1. (A)

How many unique countries per continent.

SOLUTION

```
library(dplyr)

unique_countries <- select(df, country, continent)
unique_countries <- unique(unique_countries)

print(count(group_by(unique_countries, continent)))
```

OUTPUT

```
A tibble: 6 × 2
  continent unique_countries
  <fct>      <dbl>
1 Africa          1
2 Asia            7
3 Europe         13
4 North America   3
5 Oceania         2
6 South America   4
```

QUESTION 1. (B)

Which European nation had the lowest GDP per capita in a given year?

SOLUTION

```
lowest_gdp_europe <- df %>%
  filter(continent == "Europe") %>%
  group_by(year) %>%
  select(country, year, gdpPerc) %>%
  arrange(gdpPerc) %>%
  slice(1)

print(lowest_gdp_europe)
```

OUTPUT

```
A tibble: 15 × 3
  Groups:   year [15]
  country    year gdpPerc
  <fct>      <dbl> <dbl>
1 Spain      2000  42245395
2 Spain      2001  44241012
3 Netherlands 2002  4524617
4 Netherlands 2003  4124773
5 Sweden     2004  4124775
6 Sweden     2005  4424475
7 Norway     2006  4324474
8 Norway     2007  4324538
9 Denmark    2008  4724364
10 UK        2009  4224542
11 Germany    2010  4324682
12 Finland    2011  4824264
13 France     2012  4424709
14 France     2013  4124364
15 Italy      2014  4124723
```

QUESTION 1. (c)

According to the data available, what was the average life expectancy across each continent in a given year?

SOLUTION

```
avg_life_exp <- aggregate(lifeExp ~ continent + year, data = df, FUN = mean)
print(avg_life_exp)
```

OUTPUT

```
continent year  lifeExp
1      Asia 2000 61.97577
2     Europe 2000 69.73997
3      Asia 2001 65.38429
4     Europe 2001 75.36298
5 North America 2001 77.48323
6 South America 2001 83.31628
7     Europe 2002 78.64270
8 North America 2002 72.60816
9 South America 2002 80.81569
10     Europe 2003 72.60430
11 North America 2003 80.13556
12     Oceania 2003 60.84253
13 South America 2003 60.26338
14      Asia 2004 63.39555
15     Europe 2004 82.48371
16     Oceania 2004 71.68405
17 South America 2004 76.44950
18      Asia 2005 81.82738
19     Europe 2005 69.22234
20 South America 2005 68.97537
21      Asia 2006 76.79630
22     Europe 2006 83.98541
23 South America 2006 66.72524
24      Asia 2007 72.76388
25     Europe 2007 69.83252
26     Oceania 2007 80.69767
27 South America 2007 70.18291
28      Asia 2008 73.26076
29     Europe 2008 70.36375
30     Oceania 2008 84.22953
31      Asia 2009 83.46185
32     Europe 2009 71.79938
33 North America 2009 75.90216
34      Asia 2010 66.78098
35     Europe 2010 68.22478
36 North America 2010 78.77737
37      Africa 2011 82.84786
38      Asia 2011 70.19840
39     Europe 2011 69.80194
40      Africa 2012 73.87304
41      Asia 2012 70.27315
42     Europe 2012 69.91503
43      Asia 2013 76.18017
44     Europe 2013 73.02079
45      Asia 2014 70.65546
46     Europe 2014 75.84524
```

QUESTION 1. (D)

What 5 countries have the highest total GDP over all years combined?

SOLUTION

```
gdp_df <- df %>%  
  mutate(total_gdp = as.numeric(gdpPerc) * as.numeric(pop))  
  
top_5_countries <- aggregate(total_gdp ~ country, data = gdp_df, FUN = sum) %>%  
  arrange(desc(total_gdp)) %>%  
  head(5)  
print(top_5_countries)
```

OUTPUT

	country	total_gdp
1	Russia	1.437916e+14
2	Mexico	8.652847e+13
3	Sweden	7.355758e+13
4	India	6.234923e+13
5	Netherlands	5.535939e+13

QUESTION 1. (E)

What countries and years had life expectancies of at least 80 years?

SOLUTION

```
high_life_exp <- df %>%  
  filter(lifeExp >= 80) %>%  
  select(country, year, lifeExp)  
  
print(high_life_exp)
```

OUTPUT

	country	year	lifeExp
1	USA	2003	80.13556
2	Canada	2002	80.96746
3	Japan	2005	84.46835
4	South Africa	2011	82.84786
5	Russia	2014	83.78255
6	Sweden	2004	82.48371
7	Norway	2006	83.98541
8	Argentina	2001	83.31628
9	Argentina	2002	80.81569
10	New Zealand	2007	80.69767
11	New Zealand	2008	84.22953
12	Thailand	2009	83.46185

QUESTION 1. (F)

What 10 countries have the strongest correlation (in either direction) between life expectancy and per capita GDP?

QUESTION 1. (G)

Which combinations of continent (besides Asia) and year have the highest average population across all countries?

SOLUTION

```
mutate(gdp_df, corr = lifeExp/gdpPerc)
top_10_corr <- aggregate(corr ~ country, data = gdp_df, FUN = function(x) cor(x, use =
"complete.obs")) %>%
  arrange(desc(abs(corr))) %>%

print(head(top_10_corr, 10))
```

OUTPUT

country	continent	year	lifeExp	pop	gdpPerc	total_gdp
1 India	Asia	2000	61.97577	1271264396	42184	5.362702e+13
2 India	Asia	2001	65.38429	499153698	17474	8.722212e+12
3 USA	North America	2002	64.24885	537490731	53220	2.860526e+13
4 USA	North America	2003	80.13556	246308518	15541	3.827881e+12
5 China	Asia	2004	63.39555	135614236	35285	4.785148e+12
6 China	Asia	2005	79.18640	943832654	25288	2.386764e+13
7 Brazil	South America	2006	68.16280	1199360105	6157	7.384460e+12
8 Brazil	South America	2007	70.18291	712399179	37798	2.692726e+13
9 UK	Europe	2008	74.63302	478771350	16002	7.661299e+12
10 UK	Europe	2009	68.21730	312664356	21542	6.735416e+12
11 Germany	Europe	2010	64.23460	451188843	36682	1.655051e+13
12 Germany	Europe	2011	67.17170	1096698682	9669	1.060398e+13
13 France	Europe	2012	63.65903	15747597	4709	7.415543e+10
14 France	Europe	2013	70.96158	425105885	16364	6.956433e+12
15 Italy	Europe	2014	71.83112	213793570	17723	3.789063e+12
16 Italy	Europe	2000	70.38039	1313363207	26939	3.538069e+13
17 Canada	North America	2001	77.48323	208869615	32465	6.780952e+12
18 Canada	North America	2002	80.96746	1109633463	8316	9.227712e+12
19 Australia	Oceania	2003	60.84253	70360914	12798	9.004790e+11
20 Australia	Oceania	2004	71.68405	1033056924	19899	2.055680e+13
21 Japan	Asia	2005	84.46835	655426026	53910	3.533402e+13
22 Japan	Asia	2006	76.79630	874776642	10289	9.000577e+12
23 South Korea	Asia	2007	72.76388	340780719	25249	8.604372e+12
24 South Korea	Asia	2008	73.26076	120173370	33618	4.039988e+12
25 Mexico	North America	2009	75.90216	1241919660	56900	7.066523e+13
26 Mexico	North America	2010	78.77737	1233821057	12857	1.586324e+13
27 South Africa	Africa	2011	82.84786	160204427	56702	9.083911e+12
28 South Africa	Africa	2012	73.87304	597670713	36393	2.175103e+13
29 Russia	Europe	2013	68.40803	1269887934	55844	7.091562e+13
30 Russia	Europe	2014	83.78255	1250660510	58270	7.287599e+13
31 Spain	Europe	2000	72.73875	998222050	25395	2.534985e+13
32 Spain	Europe	2001	75.36298	120919878	41012	4.959166e+12
33 Netherlands	Europe	2002	78.64270	955930590	53617	5.125413e+13
34 Netherlands	Europe	2003	72.60430	381069521	10773	4.105262e+12
35 Sweden	Europe	2004	82.48371	603973317	19775	1.194357e+13
36 Sweden	Europe	2005	69.22234	1245356331	49475	6.161400e+13
37 Norway	Europe	2006	83.98541	1281119230	30474	3.904083e+13
38 Norway	Europe	2007	69.83252	20089523	31538	6.335834e+11
39 Denmark	Europe	2008	66.09448	8011521	7364	5.899684e+10
40 Denmark	Europe	2009	75.38146	137550455	32902	4.525685e+12
41 Finland	Europe	2010	72.21495	328691868	56739	1.864965e+13
42 Finland	Europe	2011	72.43218	642156409	8264	5.306781e+12

43	Poland	Europe	2012	76.17103	901332160	43732	3.941706e+13
44	Poland	Europe	2013	79.69277	155956502	19762	3.082012e+12
45	Portugal	Europe	2014	71.92204	193366864	58526	1.131699e+13
46	Portugal	Europe	2000	66.10076	76890520	58945	4.532312e+12
47	Argentina	South America	2001	83.31628	1065119317	45590	4.855879e+13
48	Argentina	South America	2002	80.81569	1145267526	5135	5.880949e+12
49	Chile	South America	2003	60.26338	794797340	41565	3.303575e+13
50	Chile	South America	2004	76.44950	727250123	11970	8.705184e+12
51	Colombia	South America	2005	68.97537	812139894	11843	9.618173e+12
52	Colombia	South America	2006	65.28769	116880593	16960	1.982295e+12
53	New Zealand	Oceania	2007	80.69767	1213026725	14769	1.791519e+13
54	New Zealand	Oceania	2008	84.22953	865717428	17507	1.515612e+13
55	Thailand	Asia	2009	83.46185	400162831	40355	1.614857e+13
56	Thailand	Asia	2010	66.78098	519265232	40129	2.083759e+13
57	Vietnam	Asia	2011	70.19840	616056518	7588	4.674637e+12
58	Vietnam	Asia	2012	70.27315	499926939	30748	1.537175e+13
59	Philippines	Asia	2013	76.18017	29784803	45594	1.358008e+12
60	Philippines	Asia	2014	70.65546	847957709	4276	3.625867e+12

corr

1	0.001469177
2	0.003741804
3	0.001207231
4	0.005156397
5	0.001796671
6	0.003131382
7	0.011070781
8	0.001856789
9	0.004663980
10	0.003166711
11	0.001751120
12	0.006947120
13	0.013518588
14	0.004336445
15	0.004052989
16	0.002612584
17	0.002386670
18	0.009736347
19	0.004754065
20	0.003602395
21	0.001566840
22	0.007463923
23	0.002881852
24	0.002179212
25	0.001333957
26	0.006127197
27	0.001461110
28	0.002029869
29	0.001224984
30	0.001437833
31	0.002864294
32	0.001837584
33	0.001466749
34	0.006739469
35	0.004171110
36	0.001399138
37	0.002755969
38	0.002214234
39	0.008975351
40	0.002291091
41	0.001272757
42	0.008764785

```

43 0.001741769
44 0.004032627
45 0.001228890
46 0.001121397
47 0.001827512
48 0.015738207
49 0.001449859
50 0.006386758
51 0.005824147
52 0.003849510
53 0.005463990
54 0.004811192
55 0.002068191
56 0.001664157
57 0.009251239
58 0.002285454
59 0.001670838
60 0.016523728

```

QUESTION 1. (H)

Which three countries have had the most consistent population estimates (i.e. lowest standard deviation) across the years of available data?

SOLUTION

```

consistent_pop <- aggregate(pop ~ country, data = df, FUN = sd) %>%
  arrange(pop) %>%
  head(3)

print(consistent_pop)

```

OUTPUT

```

country    pop
1 Mexico  5726577
2 Russia 13595842
3  Chile  47763095

```

QUESTION 1. (I)

Excluding records from a given year, which observations indicate that the population of a country has decreased from the previous year and the life expectancy has increased?

SOLUTION

```

pop_life_change <- df %>%
  arrange(country, year) %>%
  mutate(pop_change = pop - lag(pop),
         lifeExp_change = lifeExp - lag(lifeExp)) %>%
  filter(pop_change < 0 & lifeExp_change > 0) %>%
  select(country, year, pop, lifeExp)

print(pop_life_change)

```

OUTPUT

	country	year	pop	lifeExp
1	Brazil	2007	712399179	70.18291
2	Canada	2001	208869615	77.48323
3	Chile	2004	727250123	76.44950
4	Denmark	2008	8011521	66.09448
5	India	2001	499153698	65.38429
6	Italy	2014	213793570	71.83112
7	Mexico	2010	1233821057	78.77737
8	New Zealand	2008	865717428	84.22953
9	Poland	2013	155956502	79.69277
10	Russia	2014	1250660510	83.78255
11	South Korea	2008	120173370	73.26076
12	Spain	2001	120919878	75.36298
13	Thailand	2009	400162831	83.46185
14	UK	2008	478771350	74.63302
15	USA	2003	246308518	80.13556
16	Vietnam	2012	499926939	70.27315

QUESTION 2

Create a database file "DataSet.csv" that contains 10 records of medicine with attribute : MedID, Med_Name, Company, Manf_year, Exp_date, Quantity_in_stock, Sales.

SOLUTION

```
df <- data.frame(  
  medid = 1:10,  
  med_name = c("meda", "medb", "medc", "medd", "mede", "medf", "medg", "medh", "medi",  
  "medj"),  
  company = c("comp1", "comp2", "comp1", "comp3", "comp2", "comp4", "comp3", "comp1",  
  "comp4", "comp2"),  
  manf_year = c(2018, 2019, 2018, 2020, 2019, 2021, 2020, 2018, 2021, 2019),  
  exp_date = as.Date(trimws(iconv(c("2023-12-31", "2024-06-30", "2023-11-30",  
  "2025-01-31", "2024-05-31", "2026-12-31",  
  "2025-03-31", "2023-10-31", "2026-11-30",  
  "2024-07-31"),  
    from = "", to = "UTF-8"))),  
  quantity_in_stock = c(100, 150, 200, 120, 180, 160, 140, 130, 170, 110),  
  sales = c(5000, 7000, 6000, 8000, 7500, 9000, 8500, 6500, 9500, 7200)  
)  
print(df)  
write.csv("DataSet.csv", row.names=FALSE)
```

OUTPUT

	medid	med_name	company	manf_year	exp_date	quantity_in_stock	sales
1	1	meda	comp1	2018	2023-12-31	100	5000
2	2	medb	comp2	2019	2024-06-30	150	7000
3	3	medc	comp1	2018	2023-11-30	200	6000
4	4	medd	comp3	2020	2025-01-31	120	8000
5	5	mede	comp2	2019	2024-05-31	180	7500
6	6	medf	comp4	2021	2026-12-31	160	9000
7	7	medg	comp3	2020	2025-03-31	140	8500
8	8	medh	comp1	2018	2023-10-31	130	6500
9	9	medi	comp4	2021	2026-11-30	170	9500
10	10	medj	comp2	2019	2024-07-31	110	7200

```
"x"  
"DataSet.csv"
```

QUESTION 2. (A)

Read the data file and show the first 4 record of the file.

SOLUTION

```
data <- read.csv("DataSet.csv")  
print(head(data, 4))
```

OUTPUT

MedID	Med_Name	Company	Manf_year	Exp_date	Quantity_in_stock	Sales
1	1	MedA	Comp1	2018	2023-12-31	100 5000
2	2	MedB	Comp2	2019	2024-06-30	150 7000
3	3	MedC	Comp1	2018	2023-11-30	200 6000
4	4	MedD	Comp3	2020	2025-01-31	120 8000

QUESTION 2. (B)

Read the data file and show the last 4 record of the file.

SOLUTION

```
print(tail(data, 4))
```

OUTPUT

MedID	Med_Name	Company	Manf_year	Exp_date	Quantity_in_stock	Sales
7	7	MedG	Comp3	2020	2025-03-31	140 8500
8	8	MedH	Comp1	2018	2023-10-31	130 6500
9	9	MedI	Comp4	2021	2026-11-30	170 9500
10	10	MedJ	Comp2	2019	2024-07-31	110 7200

QUESTION 2. (C)

Find the coorelation between Quantity_in_stock and Exp_date.

SOLUTION

```
data$Exp_date <- as.numeric(data$Exp_date)  
data <- na.omit(data)  
correlation <- cor(data$Quantity_in_stock, data$Exp_date)  
print(correlation)
```

OUTPUT

```
[1] 0.2084504
```

QUESTION 2. (D)

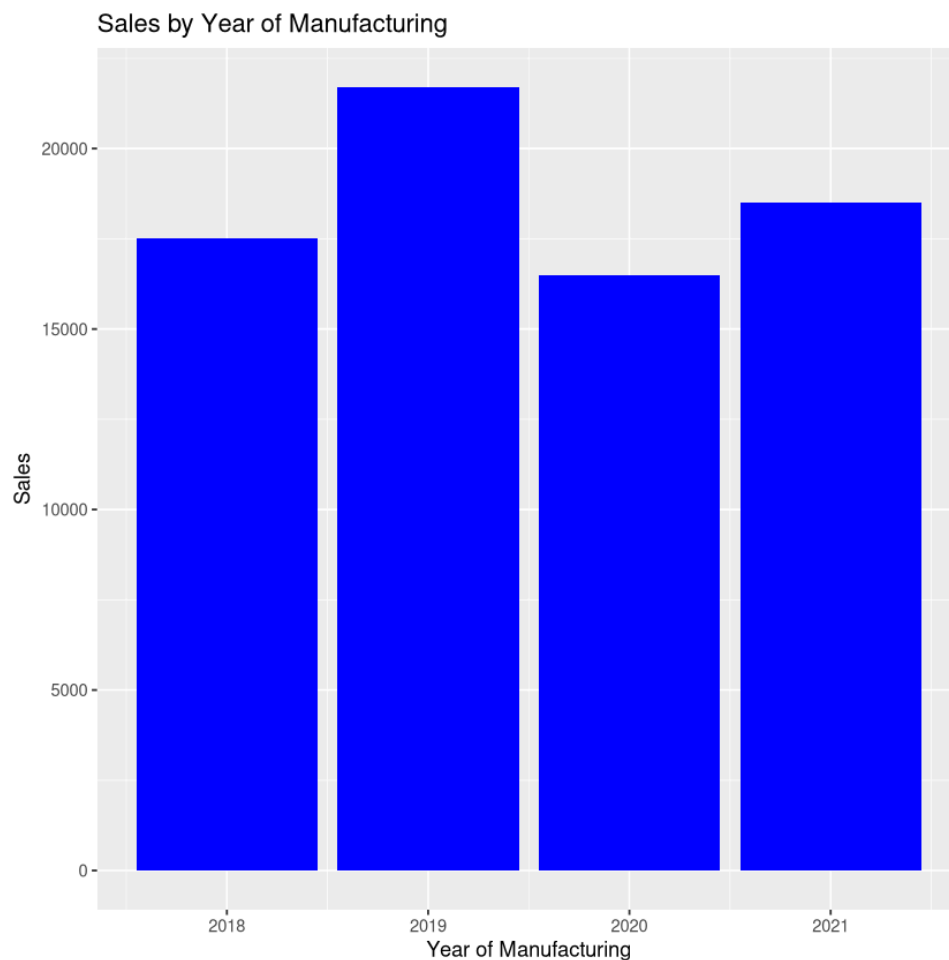
Plot the bar graph for the Sales with year of manufacturing.

SOLUTION

```
library(ggplot2)
ggplot(data, aes(x = Manf_year, y = Sales)) +
  geom_bar(stat = "identity", fill = "blue") +
  labs(
    title = "Sales by Year of Manufacturing",
    x = "Year of Manufacturing",
    y = "Sales"
  )
```

OUTPUT

plot without title



QUESTION 2. (E)

Find the company having more than one type of medicine.

SOLUTION

```
company_medicine <- aggregate(Med_Name ~ Company, data = data, FUN = function(x)
length(unique(x)))
print(company_medicine[company_medicine$Med_Name > 1, ])
```

OUTPUT

```
Company Med_Name
1   Comp1      3
2   Comp2      3
3   Comp3      2
4   Comp4      2
```

QUESTION 2. (F)

Find the type of Medicine available.

SOLUTION

```
medicine_types <- unique(data$Med_Name)
print(medicine_types)
```

OUTPUT

```
[1] "MedA" "MedB" "MedC" "MedD" "MedE" "MedF" "MedG" "MedH" "MedI" "MedJ"
```

QUESTION 2. (G)

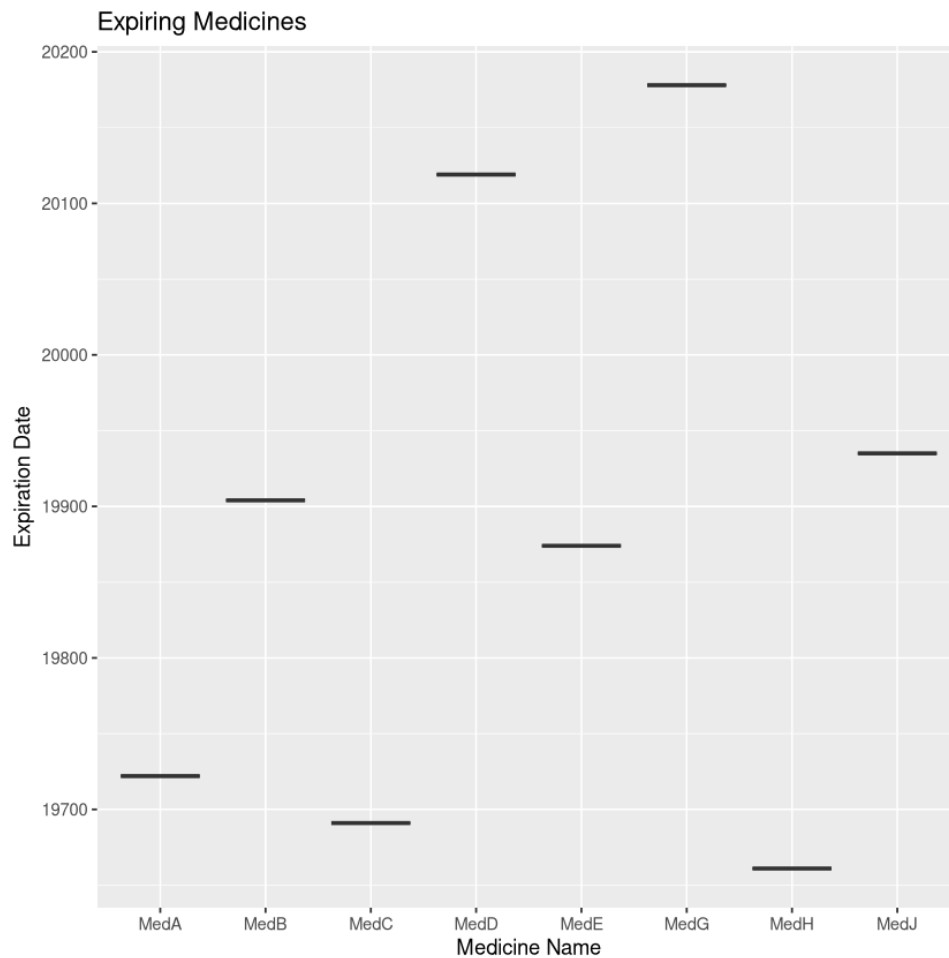
Which medicines are expiring? Show by box plots.

SOLUTION

```
expiring_medicines <- data[data$Exp_date < Sys.Date(), ]
ggplot(expiring_medicines, aes(x = Med_Name, y = Exp_date)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Expiring Medicines", x = "Medicine Name", y = "Expiration Date")
```

OUTPUT

plot without title



QUESTION 2. (H)

Find the average stock in the store.

SOLUTION

```
average_stock <- mean(data$Quantity_in_stock)
print(average_stock)
```

```
average_stock_per_med <- aggregate(Quantity_in_stock ~ Med_Name, data = data, FUN = mean)
print(average_stock_per_med)
```

OUTPUT

```
[1] 146
  Med_Name Quantity_in_stock
1    MedA             100
2    MedB             150
3    MedC             200
4    MedD             120
5    MedE             180
6    MedF             160
7    MedG             140
8    MedH             130
9    MedI             170
10   MedJ             110
```

QUESTION 2. (I)

Draw the regression line between Manufacturing year and Sales.

SOLUTION

```
ggplot(data, aes(x = Manf_year, y = Sales)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  labs(title = "Regression Line between Manufacturing Year and Sales", x = "Manufacturing  
Year", y = "Sales")
```

OUTPUT

`[[lm]] [22m`geom_smooth()` using formula = 'y ~ x'`
plot without title

